AI Paradoxes in CV Verification

SEEM5660 Agentic AI for Business and FinTech - Homework 2 Reflection

This homework turns a familiar business workflow - verifying a candidate's CV against public profiles - into an agentic AI problem. Building the verifier and then trying to fool one makes the tensions in the design visible. The central lesson is that a useful agent is not automatically a trustworthy decision-maker: its flexibility, apparent confidence, and ability to use tools can each create a new failure mode.

1. The Capability-Attack-Surface Paradox

The verifier needs tools because a language model alone cannot establish whether a CV claim is true. Searching LinkedIn, retrieving profiles, and comparing dates, schools, and skills make the system more capable. However, the CV itself becomes an attack surface. It is both the object being checked and a piece of untrusted text that may contain prompt injection, deceptive formatting, or instructions aimed at the model. Giving an agent more autonomy therefore improves its ability to investigate while also creating more ways to manipulate its reasoning.

The practical response is not to remove tools. It is to create a clear trust boundary: system instructions define the verification task, tool responses supply evidence, and the CV remains data only. A robust verifier should explicitly refuse to follow instructions embedded in a candidate document.

2. The Automation-Objectivity Paradox

A reliability score between 0 and 1 looks precise and impartial, but it compresses a chain of uncertain judgements into one number. The agent must first identify the right person among people with similar names, determine which profile fields correspond to which CV statements, decide whether two descriptions are equivalent, and finally choose a score. A score of 0.51 and a score of 0.99 have the same operational outcome under a threshold rule, even though the underlying evidence may be very different.

This suggests that high-impact KYC or recruitment decisions should preserve the evidence behind the score. The reviewer should see the selected profile, matched fields, unmatched fields, and the confidence of identity resolution. The score can prioritize work, but it should not replace accountability.

3. The Alignment-and-Data-Boundary Paradox

An agent is expected to follow the user's goal, but a malicious document can imitate the language of that goal. For example, an adversarial CV may tell an automated reviewer to treat the document as authoritative or to return a favorable assessment. If the model does not reliably distinguish higher-priority instructions from document content, following instructions becomes a route to misalignment.

The relevant control is architectural, not merely stylistic: delimit untrusted content, state that it is data, constrain tool use to approved sources, and require a structured comparison before a verdict. Instruction hierarchy is therefore a security property, not just a prompt-writing technique.

4. The Goodhart's Law Paradox

Task 2 makes Goodhart's law concrete: when a score becomes the target, a participant may optimize for the verifier rather than for truthfulness. A polished CV can omit the most checkable false field, place an embellishment in a weakly verified description, or attempt to distract the model with irrelevant text. In a real setting, this could reward candidates who understand the assessment mechanism rather than candidates with accurate credentials.

The defense is to measure what matters, not merely what is easy to measure. Structured claims such as employer, title, dates, degree, school, and skills should be compared field by field. The system should also record what it could not verify rather than silently treating missing evidence as positive evidence.

5. The Efficiency-Reliability Trade-off

Agentic verification has operational limits. More searches, cross-checks, and repeated runs can improve reliability, especially when names are ambiguous, but they consume time, API budget, and shared MCP-server capacity. A one-shot agent is faster but can be sensitive to an unlucky model decision. Repeated independent checks and median aggregation reduce instability, but they are not free.

The appropriate design depends on the decision's consequences. Low-risk screening may accept a lightweight process with human review for uncertain cases. For high-risk compliance decisions, deterministic comparisons and escalation of ambiguous identities are more appropriate than increasing model autonomy indefinitely.

Conclusion

This assignment shows that trustworthy AI is not achieved by choosing a stronger model or writing a more forceful prompt. It requires careful separation of instructions and data, grounded evidence collection, explicit uncertainty, field-level verification, and a meaningful human review path. The paradoxes do not make AI verification useless; they define the controls needed before it can be used responsibly.